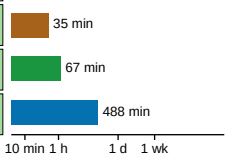
a

	Cell segmentation	Nucleus segmentation	QC & post-processing	Features	Statistics	Time to result	Scientific verdict
Published	Cellpose (default)	ilastik pixel classifier	dilation, crop, size filter	25 hand-picked	cell-level Mann–Whitney	weeks	no meaningful difference
Codex CLI	Cellpose cyto3 (half resolution)	erosion/threshold within cell	shape QC	30 features	donor-level Wilcoxon, FDR	13 min	✓ 0/30 features significant
Exploratory Agentic-J	Cellpose cyto3	Otsu threshold within cell	border, size, solidity filter	25 features	donor-level Wilcoxon, FDR	35 min	✗ 5/25 significant (3/3 runs)
Conventional Agentic-J	Cellpose cyto3 on optical density	threshold within cell	size/shape QC	32 features	donor-level permutation, FDR	67 min	✓ 0/32 significant (3/3 runs)
Conventional Agentic-J + human	Cellpose-SAM (user-specified)	micro-sam fine-tuned on 11 hand-annotated tiles	lobe-to-cell assignment	11 features	image-level Mann–Whitney, FDR	488 min	✓ 0/11 features significant

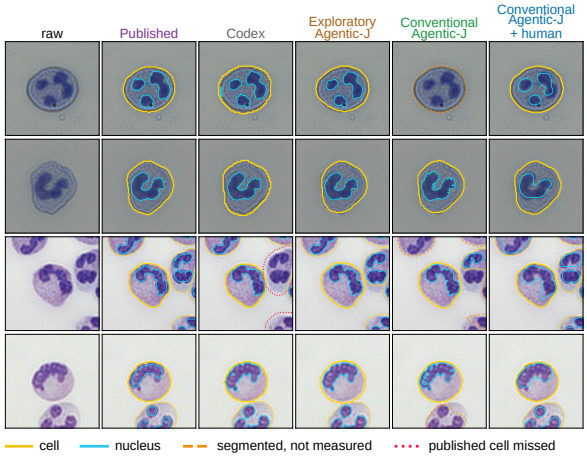


e

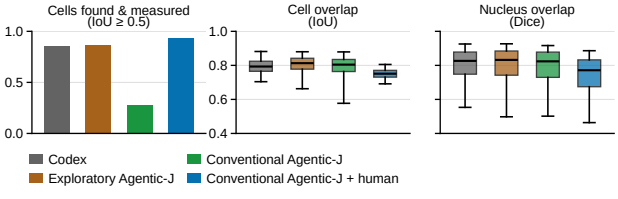
	Registration (live → fixed)	Host / nucleus segmentation	TUNEL call	Parasite localisation	Measurement & statistics	Time T1 / T2	Task 1 TUNEL calls (6 nuclei, fixed frame)	Task 2 proliferation index (15 parasites, last live frame)
Published	by hand	hand-drawn ROIs	by inspection	15 ROIs + 3 backgrounds each, by hand	ROI Manager → Excel; Mann–Whitney	not recorded	2/6 nuclei TUNEL+	8 TUNEL+ vs 7 TUNEL−, p = 0.0012
Codex CLI	SIFT + RANSAC, ECC refined (1.6 px)	T1 DAPI threshold; T2 Cellpose cyto3	largest-gap cutoff	red-channel peaks + water-shed: 10 kept	annulus background; no test	18 min (both)	✓ 6/6 calls, r = 0.98	✓ 10/15 found, hosts right; p = 0.017
Exploratory Agentic-J	TurboReg rigid on brightfield	T1 StarDist; T2 Cellpose cyto3 (7 hosts)	largest-gap cutoff	C2 + C3 spot evidence: 22 kept	annulus background; no test	22 / 69 min	✓ 5/6 calls, r = 0.95, 1 split	✗ 0/15 found (+22), no host
Conventional Agentic-J	TurboReg / StackReg rigid	T1 DoG + water-shed; T2 Cellpose cyto3 (6 hosts)	2-component GMM	micro-sam vit_b_lm: 105 → 5 kept	annulus background; no test	55 / 85 min	✓ 5/6 calls, r = 0.99	✗ 3/15 found (+2), no TUNEL+
Conventional Agentic-J + human	lab ROIs given, origin translated	lab ROIs (given)	largest-gap cutoff	lab ROIs paired 1:1 with backgrounds	native 16-bit; Mann–Whitney	32 min / n.r.	✓ 5/6 calls, r = 0.97	✓ 15/15; 14/15 within 0.1; p = 0.0012

Manual (analyst) · Human feedback (user) · Automatic (Agentic-J) · Automatic (other agent) · ✓ published conclusion reproduced · ✗ not reproduced

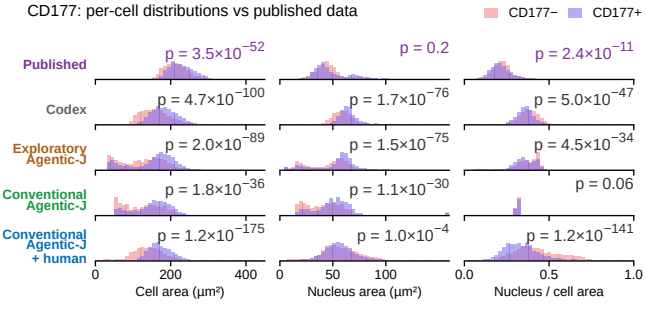
b CD177: segmentations of four example cells



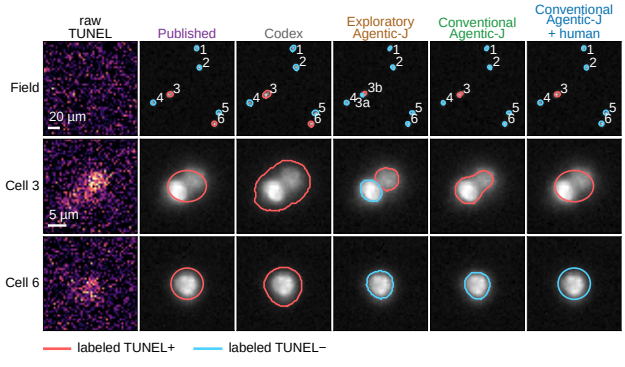
c CD177: agreement with the published masks (4389 cells)



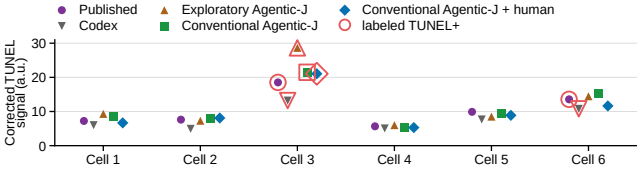
d CD177: per-cell distributions vs published data



f Leishmania task 1: six nuclei, called by each pipeline



g Leishmania task 1: corrected TUNEL signal per nucleus



h Leishmania: agreement with the published analysis (counts)

		Codex CLI	Exploratory Agentic-J	Conventional Agentic-J	Conventional Agentic-J + human
Task 1	TUNEL calls agree	6 / 6	5 / 6	5 / 6	5 / 6
Task 2	Lab parasites found	10 / 15	0 / 15	3 / 15	15 / 15
	Host group correct	10 / 10	–	2 / 3	15 / 15
	Index within 0.1 of lab	4 / 10	–	1 / 3	14 / 15
	False detections	0	22	2	0

i Leishmania task 2: proliferation index by host TUNEL status

